TurboQuant v1.4.0

Work Log

2026-04-02 ~ 04-03

GLM-4.7-Flash + V Cache Bug Fix

head_dim=576 Block Split Implementation

IWHT FP16→FP32 정밀도 Fixed

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Session Overview

- 04-02 PM — GLM-4.7-Flash head_dim=576 design review + implementation plan
- 04-02 PM — Type system + block structs + auto-mapping + quantization kernels
- 04-02 PM — K dot product + V dequant + Q preprocessing + IWHT
- 04-02 PM — Template dispatch + 20 instance files + build system
- 04-02 EVE — Build success + GLM-4.7-Flash generation test (49.4/65.6 t/s)
- 04-02 EVE — GitHub Issue #2 bug reproduction (Windows + Linux)
- 04-02 EVE — V cache IWHT FP16 precision bug found + FP32 fix
- 04-03 AM — Q preprocessing WHT also FP32 (ROCm compat)
- 04-03 AM — v1.4.0 release + workflow + community comments

GLM-4.7-Flash head_dim=576 Core

WHT supports power-of-2 only → 576 = **256 + 256 + 64** Block 분할로 Fixed

BLOCK	RANGE	WHT	CORRECTION	KERNEL
Sub-block 1	[0, 256)	WHT_256	QJL SRHT	Reuses existing _0
Sub-block 2	[256, 512)	WHT_256	QJL SRHT	Reuses existing _0
Sub-block 3	[512, 576)	WHT_64	Direct Sign	Reuses existing _2

Simulation result: MSE -3.4% vs Full QR orthogonal rotation, memory overhead 0%. 256이 sweet spot — 512 한 덩어리보다 256 두 개가 MSE 낮음.

Simulation Comparison (500회 반복, 3-bit) Verified

METHOD	MSE	VS FULL QR	STORAGE	OVERHEAD
FullPad 1024	0.063	-40.6%	1024	78%
256+256+64	**0.102**	**-3.4%**	**576**	**0%**
Full QR ortho	0.106	기준	576	0%
512+64	0.108	+2.5%	576	0%
Kronecker H64⊗O9	0.112	+5.2%	576	0%
No rotation	0.569	+435%	576	0%

Implementation Details 32 files

TYPE SYSTEM

- 4 enum: TBQ3_4 ~ TBQP4_4 (ID 57-60)
- 4 composite block struct (228-300B)
- blck_size = 576, type_traits 등록

QUANTIZATION (ENCODING)

- 4 quantize 함수 (sub-block별 독립 WHT)
- Sub-block 1,2: QJL SRHT Correction
- Sub-block 3: Direct Sign Correction

DEQUANTIZATION (DECODING)

- K dot product 4개 + V dequant 2개
- Q 전처리: 3-pass split WHT
- V IWHT: 3-pass 역변환 → dst

INFRASTRUCTURE

- 20 template instance (.cu) 파일
- CMakeLists + fattn.cu dispatch
- auto-mapping head_dim=576 → _4

V Cache IWHT FP16 Precision Bug Critical

SYMPTOMS

- ~1500-2300토큰 이후 `////` 또는 `???` 반복
- tbqp3: ~1826tok에서 발생
- tbqp4: ~2284tok에서 발생
- f16 V: 문제 없음

ROOT CAUSE

- V IWHT butterfly가 FP16(half)으로 수행
- Accumulated WHT-domain values → FP16 exceeded precision
- Catastrophic cancellation in subtraction
- K-only quant → OK, V-only quant → Corrupted

K/V 분리 테스트가 Core: tbqp3 K + f16 V = OK / f16 K + tbq3 V = Corrupted → V Path 특정

Fix: All WHT/IWHT Converted to FP32 Fixed

PATH	BEFORE	AFTER	IMPACT
V IWHT (D=256/128/64/576)	FP16 butterfly	FP32 butterfly	corruption Fixed
Q WHT MSE (D=256/128/64/576)	FP16 butterfly	FP32 butterfly	Precision + ROCm
Q WHT QJL (D=256/576)	FP16 butterfly	FP32 butterfly	Precision + ROCm

VERIFIED 결과

Qwen3.5-27B tbqp3/tbq3	4096 tok	NO CORRUPTION
파울리 테스트	볼프강 파울리	Correct
하이젠베르크 테스트	베르너 하이젠베르크	Correct

SIDE EFFECTS

- KV 캐시 압축률: Unchanged
- 속도 Overhead: Negligible
- ROCm/HIP 빌드 호환성 개선
- `__shfl_xor_sync` half → float

Community Issues Done

ISSUE	REPORTER	DESCRIPTION	STATUS
#2	@alrunan	//// 출력 (RTX 5090, Qwen3.5-27B)	**v1.4.0 Fixed**
#2	@superherointj	Linux NixOS에서도 동일 재현	**v1.4.0 Fixed**
#3	@jarkevithwlad	standalone 크래시 (RTX 3090)	v1.4.0 Requested to try
#4	@dpblnt	Vulkan SET_ROWS 미지원	CUDA-only notice
#5	@dpblnt	ROCm(HIP) 빌드 실패	**FP32 전환으로 Fixed**
#6	@jarkevithwlad	Gemma 4 아키텍처 미인식	Needs upstream

v1.4.0 Release Released

32	4	0
Fixed/추가 파일	New Types (TBQ_4)	Corruption (4096tok)

- GitHub Release: v1.4.0 생성 + 릴리즈 노트
- Build workflow triggered (Linux x64/arm64, Windows x64, CUDA 12.8)
- README.md updated (Korean + English)
- Issue #2, #3, #4, #5, #6 comments responded
- llama.cpp Discussion #20969 announcement
- turboquant/ directory backup (modified_files + patches)